

Product Requirements Document (PRD)

Mirror

Version

1.0 MVP

Owner

AI Engineering Internship Project

Product Type

AI Personality & Communication Style Cloning System

1. Executive Summary

Mirror creates a digital communication profile of a user and generates text and voice outputs that resemble the user's communication style.

The system learns from:

- Emails
- Messages
- Documents
- Transcripts
- Voice recordings

Mirror does not simply imitate vocabulary.

It attempts to model:

- Tone
- Formality
- Directness
- Humor
- Sentence structure
- Communication habits

- Speaking patterns

The goal is:

"If someone receives the generated message, they should believe the user wrote or spoke it."

2. Problem Statement

People communicate in unique ways.

Current AI assistants generate generic responses.

Problems:

- Messages do not sound like the user
- Brand voice becomes inconsistent
- Executives spend time editing AI output
- Teams struggle to maintain communication style

Users want AI-generated content that sounds authentically like them.

3. Product Vision

Create a personalized AI communication engine capable of:

- Learning a person's style
- Replicating their writing
- Replicating their speaking style
- Measuring similarity objectively

The system should answer:

"How would I say this?"

instead of:

"How would AI say this?"

4. Goals

Primary Goals

Allow users to:

- Create style profiles
 - Generate text in their style
 - Generate voice responses in their style
 - Evaluate similarity
-

Secondary Goals

Provide:

- Style analytics
 - Communication insights
 - Persona comparison tools
-

5. Non Goals

The MVP will NOT include:

- Real-time voice calls
 - Video generation
 - Face cloning
 - Social media automation
 - Autonomous posting
 - Team personas
-

6. Target Users

Content Creators

Need consistent voice.

Executives

Need personalized communication.

Founders

Need scalable communication.

Professionals

Want AI-generated content that sounds authentic.

7. Success Metrics

Product Metrics

Profile Completion Rate

Target:
90%

Text Generation Usage

Target:
70%

Voice Generation Usage

Target:
50%

AI Metrics

Text Similarity Score

Target:
80%

Voice Similarity Score

Target:
80%

Blind Test Success Rate

Target:
60%

Definition:

Users cannot reliably distinguish between real and generated content.

8. User Stories

US-1

As a user,

I want to upload my writing samples,

so Mirror can learn my style.

US-2

As a user,

I want to upload voice recordings,

so Mirror can learn how I speak.

US-3

As a user,
I want to generate messages,
so they sound like me.

US-4

As a user,
I want voice outputs,
so the generated speech resembles my speaking style.

US-5

As a user,
I want similarity scores,
so I know how accurate the profile is.

9. Functional Requirements

FR-1 Authentication

User can:

- Register
- Login
- Logout

JWT authentication.

FR-2 Text Sample Upload

Supported formats:

- TXT
- PDF
- DOCX

System stores samples.

FR-3 Voice Sample Upload

Supported formats:

- MP3
- WAV
- M4A

System stores recordings.

FR-4 Audio Transcription

Convert audio into text.

Store:

- transcript
 - timestamps
-

FR-5 Style Profile Creation

Generate a style profile.

Attributes:

- Formality
- Directness
- Humor
- Verbosity
- Positivity
- Technicality

Example:

Directness: 0.9

Humor: 0.3

Verbosity: 0.6

FR-6 Voice Profile Creation

Generate voice profile.

Attributes:

- Speaking rate
 - Pause frequency
 - Average response length
 - Conversational tone
-

FR-7 Text Generation

User enters:

Prompt

Example:

"Write a meeting cancellation email."

Mirror generates content using user profile.

FR-8 Voice Generation

User enters:

Prompt

Mirror generates:

- text
 - audio version
-

FR-9 Style Consistency Engine

Every generation must use:

- style profile
- examples
- communication rules

Output should remain consistent.

FR-10 Similarity Evaluation

Compare:

Generated content

vs

Training samples

Produce:

Similarity Score

FR-11 Blind Test Mode

Display:

Sample A

Sample B

User guesses:

Human or AI

Track accuracy.

FR-12 Profile Retraining

User can upload additional samples.

Profile updates automatically.

10. AI Processing Pipeline

Step 1

Data Collection

Input:

Documents

Messages

Audio

Step 2

Transcription

Audio → Text

Step 3

Style Extraction

Extract:

- vocabulary patterns
 - sentence length
 - tone
 - formality
 - communication habits
-

Step 4

Profile Generation

Create structured profile.

Step 5

Retrieval

Select representative examples.

Step 6

Generation

Use:

Prompt

-

Profile

-

Examples

↓

Generate response

Step 7

Evaluation

Compare generated output against profile.

Generate similarity score.

11. Style Profile Schema

Text Profile

formality

directness

verbosity

humor

technicality

positivity

sentence_length

question_frequency

emoji_usage

Voice Profile

speaking_rate

pause_frequency

average_sentence_duration

energy_level

conversation_style

response_length

12. Database Design

Users

id

email

password_hash

created_at

Documents

id

user_id

file_path

file_type

created_at

AudioFiles

id

user_id

file_path

duration

created_at

Transcripts

id

audio_id

content

created_at

StyleProfiles

id

user_id
profile_json
created_at
updated_at

Generations

id
user_id
prompt
output_text
similarity_score
created_at

VoiceGenerations

id
generation_id
audio_path
created_at

13. Technical Architecture

Frontend

NiceGUI

Responsibilities:

- Upload interface

- Profile dashboard
 - Text generation
 - Voice generation
 - Evaluation screens
-

Backend

FastAPI

Responsibilities:

- Authentication
 - Profile generation
 - AI orchestration
-

Database

PostgreSQL

Responsibilities:

- User data
 - Profiles
 - Generations
-

Workers

Celery

Responsibilities:

- Transcription
 - Profile building
 - Evaluation jobs
-

Deployment

Docker

Render

14. UI Requirements

Login Page

Email

Password

Register

Login

Profile Dashboard

Cards:

Documents Uploaded

Audio Uploaded

Profile Strength

Similarity Score

Upload Page

Upload:

- Documents
- Audio

Processing status visible.

Style Analysis Page

Display:

Directness

Formality

Humor

Verbosity

Technicality

Visualized with charts.

Text Generation Page

Input:

Prompt

Output:

Generated text

Similarity score

Voice Generation Page

Input:

Prompt

Output:

Generated text

Generated audio

Play button

Download button

Evaluation Page

Display:

Generated samples

Historical samples

Similarity metrics

Blind test interface

15. NiceGUI Requirements

Must use:

`ui.upload()`

`ui.card()`

`ui.table()`

`ui.dialog()`

`ui.notify()`

`ui.audio()`

`ui.echart()`

Responsive layout required.

16. Security Requirements

JWT authentication

Password hashing

Private profile access

File validation

Rate limiting

Audit logging

Encrypted storage for uploads

17. Evaluation Framework

Metric 1

Text Similarity

Target:
80%

Metric 2

Voice Similarity

Target:
80%

Metric 3

Profile Consistency

Target:
85%

Question:

Does the style remain stable across generations?

Metric 4

Blind Test Accuracy

Target:
60%

If users cannot consistently identify AI-generated content, the profile is effective.

Metric 5

Generation Quality

Target:
4/5 user rating

18. Deliverables

Source Code

Dockerfile

docker-compose.yml

Database Migrations

README

Architecture Diagram

ER Diagram

Prompt Design Documentation

Evaluation Report

Demo Video

19. Acceptance Criteria

User can:

- ✓ Register
- ✓ Login
- ✓ Upload text samples
- ✓ Upload voice samples
- ✓ Generate style profile

- ✓ Generate text in learned style
 - ✓ Generate voice output
 - ✓ View profile analytics
 - ✓ View similarity scores
 - ✓ Run blind tests
 - ✓ Deploy on Render
 - ✓ Use NiceGUI
 - ✓ Use PostgreSQL
 - ✓ Use Docker
 - ✓ Use FastAPI
-

20. Stretch Goals

Multi-Person Personas

Team Voice Profiles

Historical Personality Evolution

Meeting Participation

Live Voice Conversation Mode

Persona Comparison Engine

Brand Voice Management